

6 Tritium is an isotope of hydrogen and is represented by the symbol ${}^3_1\text{H}$.

(a) Explain the term *isotope*.

.....
..... [1]

(b) Tritium has a binding energy per nucleon of 2.83 MeV.

(i) Explain what is meant by the *binding energy* of a nucleus.

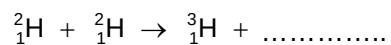
.....
..... [1]

(ii) Given that the mass of a proton is $1.00783\text{ }u$ and the mass of a neutron is $1.00867\text{ }u$, calculate, to five decimal places, the mass of a tritium nucleus.

mass = u [3]

(c) In a fusion power reactor, two deuterium nuclei fuse together to form a tritium nucleus.

(i) Complete the nuclear equation below for the fusion of deuterium.



[1]

(ii) For the fusion of deuterium shown in **(i)**, 4.03 MeV of energy is released.

Calculate

1. the binding energy of a deuterium nucleus,

binding energy = MeV [2]

2. the mass of deuterium required per unit time to produce 2.86×10^9 W of power.

mass = kg [3]

7. The decay of radioactive materials is a random process. On average, nuclides which